CSC 345 – Project #3 AI Essay

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For Project 3, we used AI tools in two separate capacities throughout the development of the virtual memory manager. Github Copilot was used as a primary debugging tool to fix errors during the development of main2 and main3. Once a scaffolding of functions with instruction inline comments were established manually, Gemini was asked to implement each function, one at a time. This was to ensure the scope of the changes could be tracked and that there were no deviations that would impact the final result. The combination of human ideation and small, boiler plate code led to a very successful first deployment of main2 and main3. As a last step, Gemini was asked to check the commented code for accuracy, and if it met the previously stated design intent. In addition, during the project Ben's SSD failed and required another development environment setup. Setting up WSL and Ubuntu on the computer was another task that utilized AI assistance. Overall, we again found AI tools helpful for accelerating debugging, understanding concepts, and presenting a solid "sounding board" for ideas. However, these tools were not able to implement the Project 3 as fully desired. Many suggestions were not feasible inside the scope of our project. For example, Gemini wanted to compose a test script that would try to recompile main2 and main3 for the data collection portion of the Project. Again, the most effective workflow was iterative: write code, test it, analyze behavior, consult AI for clarification, then re-test. We observed a drastic time cost for this project by utilizing AI in main2 and main3, however, validation, critical thinking, and manual testing remained essential throughout the project.